clusterOne: Pearson vs sMAPE

A Spotlight — Side-Deck for Week of April 30

Binghamton University

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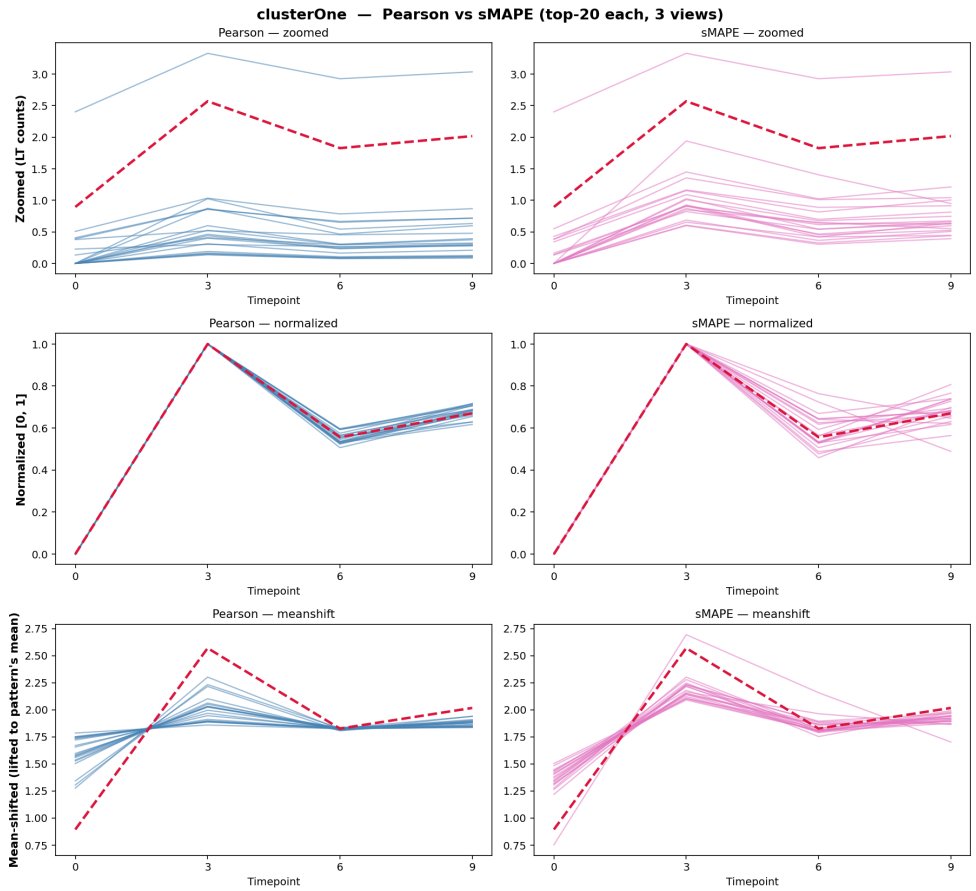
Why This Question

- Main deck picked **Pearson** for clusterOne (rawMAE failed shape; fall back to principled scale-invariant shape metric).
- But sMAPE also passes the dual gate cleanly on clusterOne and looks slightly tighter to the pattern in the mean-shift view.
- Both score **shape** $= 1.00$ (every top-20 gene has $r > 0.9$ vs pattern). Both pass the magnitude gate too. Need a tiebreaker.
- This deck: visual + numerical comparison so the pick is defended, not assumed.

Verdict (preview): Pearson, but for sharper reasons than the main deck stated.

The 3-View Comparison



What the Three Views Say

- **Zoomed:** Pearson genes sit at LT $\sim$ 0–1; sMAPE genes at LT $\sim$ 1–2. sMAPE is visibly closer to the pattern's actual range (0.85–2.55).
- **Normalized:** Pearson shows a tight overlay on the pattern's shape. sMAPE shows visible fanning, especially after t=3.
- **Mean-shifted:** sMAPE traces the up-down-up spike better; Pearson genes look flatter around the pattern's mean.

So which view do we trust? Different views answer different questions. The next two slides ground each in a number so it's not eyeballing.

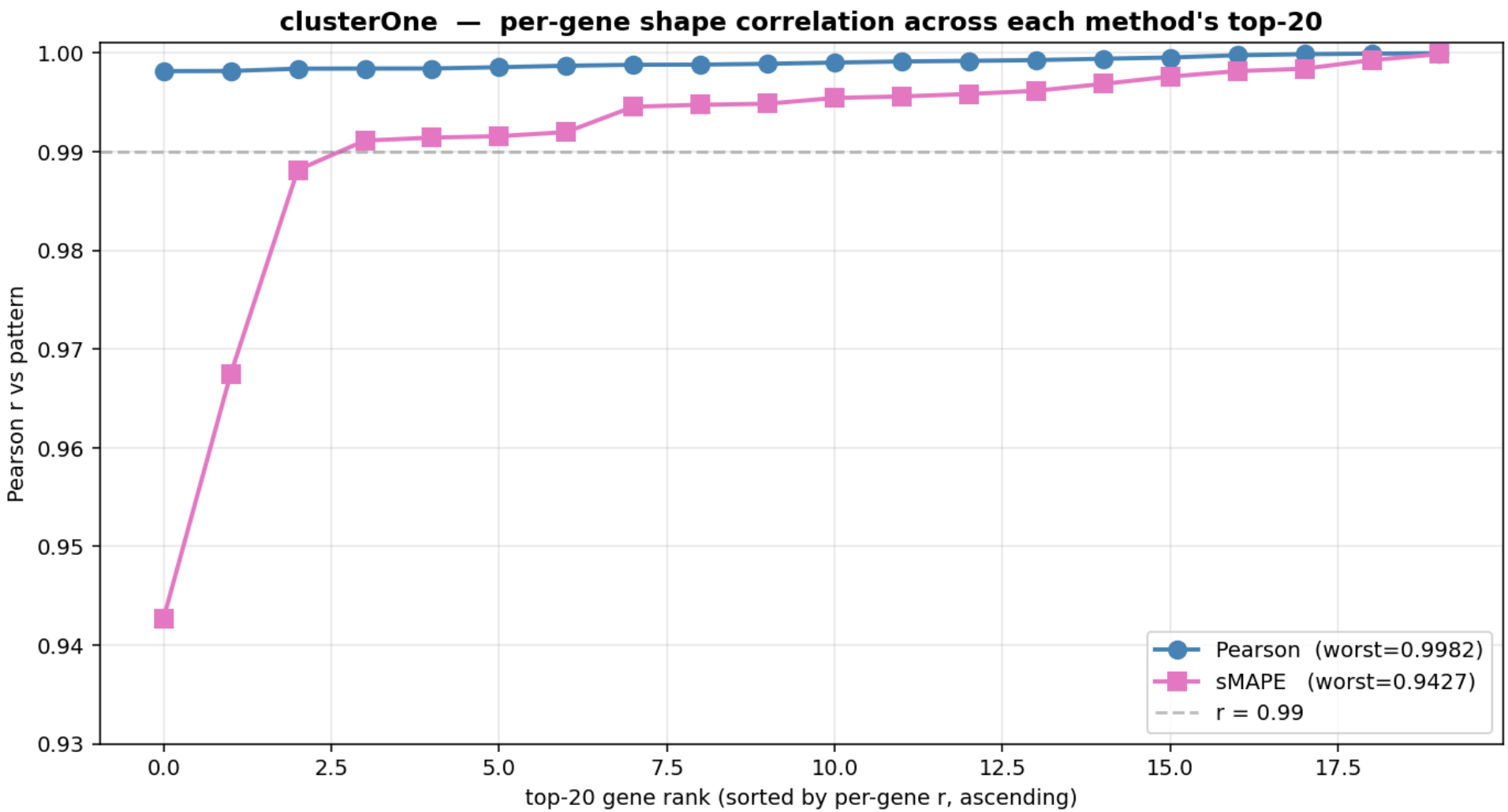
Number 1 — Amplitude (Why sMAPE Wins Mean-Shift)

Mean dynamic range (max − min) of each method's top-20 on clusterOne:

Method	**Mean LT range**
Pearson	0.41
sMAPE	**0.87** (∼2× Pearson)
Pattern itself	*1.68*

sMAPE's picks are at **52% of the pattern's amplitude**; Pearson's are at **25%**. ⇒ The mean-shift advantage is **real**, not an artifact. sMAPE picks higher-amplitude shape-correct genes that more closely trace the pattern's spike when shifted to the same mean.

Number 2 — Shape Consistency (Why Pearson Wins Normalized)



The Per-Gene r Distribution Tells the Whole Story

Method	**Worst** r	**Median** r	$\# r > 0.99$	$\# r > 0.998$
Pearson	**0.998**	0.999	**20 / 20**	**20 / 20**
sMAPE	0.943	0.995	17 / 20	0 / 20

- **Pearson**: every one of the 20 picks has $r \geq 0.998$. Essentially perfect shape correlation, top to bottom.
- **sMAPE**: median is comparable, but the bottom 3 of 20 fall below $r = 0.99$, with the worst at 0.94. Some shape variance is being absorbed.

So the trade is sharp: sMAPE buys $\sim 2\times$ amplitude proximity by accepting noticeable shape variance on the bottom of its top-20. Pearson keeps shape extremely tight at the cost of amplitude.

The Verdict

The trade-off:

	sMAPE buys	**Pearson buys**
Amplitude proximity	**52% of pattern's range**	25%
Worst-case shape	0.94	**0.998**
Empirical consensus	4 / 20 with Pearson	— (set the consensus)

The prof ruled shape > magnitude. Under that priority:

- sMAPE's amplitude advantage is *good but secondary*.
- sMAPE's shape variance on the bottom genes is *bad and primary*.

Pick: Pearson. Trade goes the wrong direction for sMAPE — it accepts shape compromise to gain magnitude proximity, but magnitude isn't what we're prioritizing.

If the prof's priority were reversed (magnitude over shape), this would flip — sMAPE would be the right pick.

Side Note — Why Our Shape-Test Saturated

The dual gate's **shape axis** = fraction of top-20 with $r > 0.9$. Both Pearson and sMAPE score 1.00 on it because every gene clears the 0.9 bar. But the per-gene r distribution is much wider on sMAPE (worst = 0.94) than on Pearson (worst = 0.998). The 0.9 threshold was useful as a first cut but too lenient to differentiate fine-grained shape consistency. **Practical fix:** for tie-breaks, look at the worst-case r, not the fraction-above-threshold. Pearson's worst gene is at $r = 0.998$ — Fréchet's, MSE's, and Pearson's are all in this $r > 0.998$ regime. sMAPE and Cosine are not.